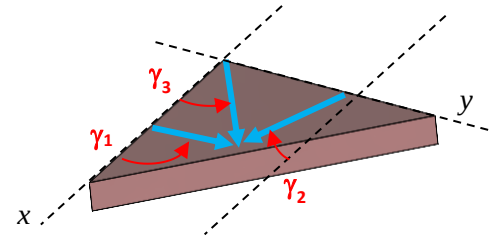
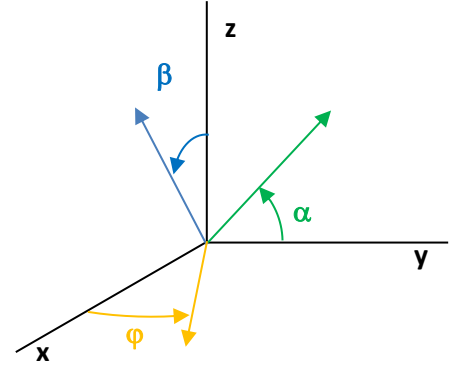
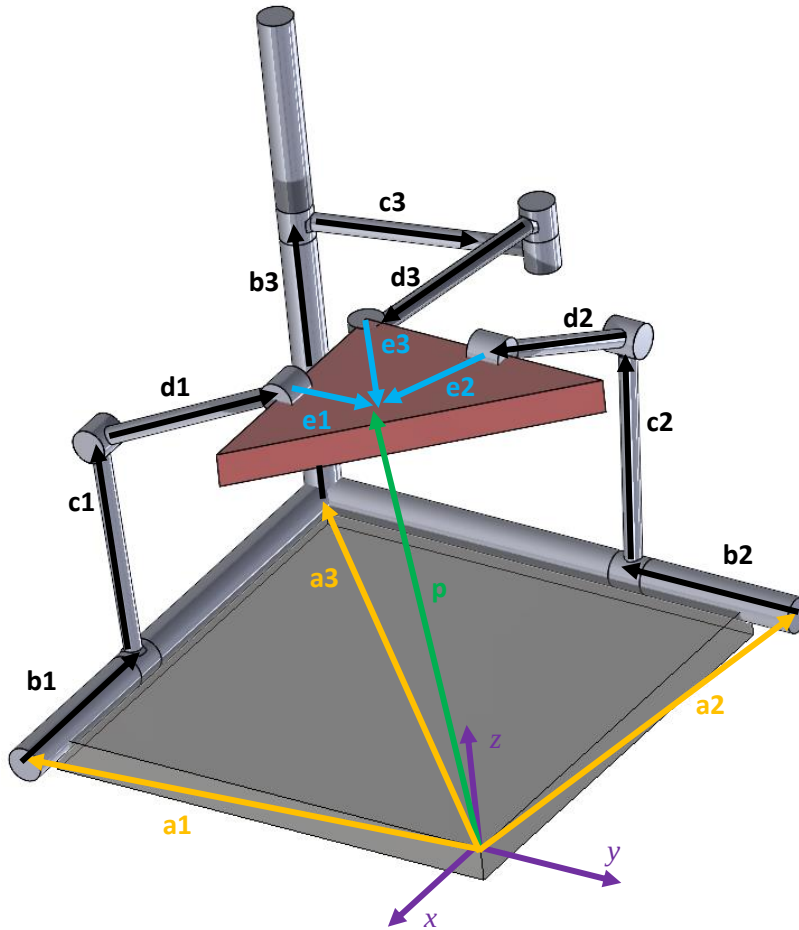


9. Tripteron



$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 + \mathbf{c}_1 + \mathbf{d}_1 + \mathbf{e}_1 - \mathbf{p} \\ \mathbf{a}_2 + \mathbf{b}_2 + \mathbf{c}_2 + \mathbf{d}_2 + \mathbf{e}_2 - \mathbf{p} \\ \mathbf{a}_3 + \mathbf{b}_3 + \mathbf{c}_3 + \mathbf{d}_3 + \mathbf{e}_3 - \mathbf{p} \end{bmatrix}$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{x1} + b_1 + 0 + 0 + e_1 \cos(\gamma_1) - p_x \\ a_{y1} + 0 + c_1 \cos(\alpha_c) + d_1 \cos(\alpha_d) + e_1 \sin(\gamma_1) - p_y \\ a_{z1} + 0 + c_1 \sin(\alpha_c) + d_1 \sin(\alpha_d) + e_1 - p_z \\ a_{x2} + 0 + c_2 \sin(\beta_c) + d_2 \sin(\beta_d) + e_2 \cos(\gamma_2) - p_x \\ a_{y2} + b_2 + 0 + 0 + e_2 \sin(\gamma_2) - p_y \\ a_{z2} + 0 + c_2 \cos(\beta_c) + d_2 \cos(\beta_d) + e_{2z} - p_z \\ a_{x3} + 0 + c_3 \cos(\varphi_c) + d_3 \cos(\varphi_d) + e_3 \cos(\gamma_3) - p_x \\ a_{y3} + 0 + c_3 \sin(\varphi_c) + d_3 \sin(\varphi_d) + e_3 \sin(\gamma_3) - p_y \\ a_{z3} + b_3 + 0 + 0 + e_{3z} - p_z \end{bmatrix}$$

$$\mathbf{q} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \alpha_c \\ \alpha_d \\ \beta_c \\ \beta_d \\ \varphi_c \\ \varphi_d \end{bmatrix}$$

We add the driving constraints,

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 + \mathbf{c}_1 + \mathbf{d}_1 + \mathbf{e}_1 - \mathbf{p} \\ \mathbf{a}_2 + \mathbf{b}_2 + \mathbf{c}_2 + \mathbf{d}_2 + \mathbf{e}_2 - \mathbf{p} \\ \mathbf{a}_3 + \mathbf{b}_3 + \mathbf{c}_3 + \mathbf{d}_3 + \mathbf{e}_3 - \mathbf{p} \\ p_x - f_1(t) \\ p_y - f_2(t) \\ p_z - f_3(t) \end{bmatrix}$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{x1} + b_1 + e_1 \cos(\gamma_1) - p_x \\ a_{y1} + c_1 \cos(\alpha_c) + d_1 \cos(\alpha_d) + e_1 \sin(\gamma_1) - p_y \\ a_{z1} + c_1 \sin(\alpha_c) + d_1 \sin(\alpha_d) + e_{1z} - p_z \\ a_{x2} + c_2 \sin(\beta_c) + d_2 \sin(\beta_d) + e_2 \cos(\gamma_2) - p_x \\ a_{y2} + b_2 + e_2 \sin(\gamma_2) - p_y \\ a_{z2} + c_2 \cos(\beta_c) + d_2 \cos(\beta_d) + e_{2z} - p_z \\ a_{x3} + c_3 \cos(\varphi_c) + d_3 \cos(\varphi_d) + e_3 \cos(\gamma_3) - p_x \\ a_{y3} + c_3 \sin(\varphi_c) + d_3 \sin(\varphi_d) + e_3 \sin(\gamma_3) - p_y \\ a_{z3} + b_3 + e_{3z} - p_z \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \alpha_c \\ \alpha_d \\ \beta_c \\ \beta_d \\ \varphi_c \\ \varphi_d \\ p_x \\ p_y \\ p_z \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -c_1 \sin(\alpha_c) & -d_1 \sin(\alpha_d) & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & c_1 \cos(\alpha_c) & d_1 \cos(\alpha_d) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & c_2 \cos(\beta_c) & d_2 \cos(\beta_d) & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -c_2 \sin(\beta_c) & -d_2 \sin(\beta_d) & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_3 \sin(\varphi_c) & -d_3 \sin(\varphi_d) & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_3 \cos(\varphi_c) & d_3 \cos(\varphi_d) & 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{j} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c_1\dot{\alpha}_c \cos(\alpha_c) & -d_1\dot{\alpha}_d \cos(\alpha_d) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c_1\dot{\alpha}_c \sin(\alpha_c) & -d_1\dot{\alpha}_d \sin(\alpha_d) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -c_2\dot{\beta}_c \sin(\beta_c) & -d_2\dot{\beta}_d \sin(\beta_d) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -c_2\dot{\beta}_c \cos(\beta_c) & -d_2\dot{\beta}_d \cos(\beta_d) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_3\dot{\phi}_c \cos(\phi_c) & -d_3\dot{\phi}_d \cos(\phi_d) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c_3\dot{\phi}_c \sin(\phi_c) & -d_3\dot{\phi}_d \sin(\phi_d) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{\dot{q}} = \begin{bmatrix} \dot{b}_1 \\ \dot{b}_2 \\ \dot{b}_3 \\ \dot{\alpha}_c \\ \dot{\alpha}_d \\ \dot{\beta}_c \\ \dot{\beta}_d \\ \dot{\phi}_c \\ \dot{\phi}_d \\ \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{bmatrix} \quad \mathbf{\ddot{q}} = \begin{bmatrix} \ddot{b}_1 \\ \ddot{b}_2 \\ \ddot{b}_3 \\ \ddot{\alpha}_c \\ \ddot{\alpha}_d \\ \ddot{\beta}_c \\ \ddot{\beta}_d \\ \ddot{\phi}_c \\ \ddot{\phi}_d \\ \ddot{p}_x \\ \ddot{p}_y \\ \ddot{p}_z \end{bmatrix} \quad \dot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\dot{p}_x^0 - \ddot{p}_x^0 t \\ -\dot{p}_y^0 - \ddot{p}_y^0 t \\ -\dot{p}_z^0 - \ddot{p}_z^0 t \end{bmatrix} \quad \ddot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\ddot{p}_x^0 \\ -\ddot{p}_y^0 \\ -\ddot{p}_z^0 \end{bmatrix}$$

Keep in mind that the angles γ_i are the angles of the vectors at the end effector from the R joint to the point of interest (Pol). The angle is measure on the xy plane with zero value coinciding with the x axis.

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% This code solves the kinematic analysis of the Tripteron linkage using multibody kinematics based on the use of close loop equation
% c,d:      length of the links
% a:        location of the origin for the linear actuators
% b:        stroke of the linear actuators
% e:        location of the Pol with respect the connecting R joints at the effector
% q:        vector with the kinematic variables
% P:        Matrix with the historic position
% V:        Matrix with the historic velocities
% A:        Matrix with the historic acceleration
% Fq:       Vector with kinematic constraints
% J:        Jacobian matrix
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all
clc
close all
%vector from gcs to origin of sliders
a1=[0; -300; 0]; a2=[-300; 0; 0]; a3=[-300; -300; 0];
%length of first link
c=150;
%length of second link
d=150;
%distance of the effector Pol to the effector joints
e1=[0; 90; 0]; e2=[90; 0; 0]; e3=[90; 90; 0];
%driving constraints
px0=-2; vx0=-1; ax0=0;
py0=-3; vy0=-1; ay0=0;
pz0=4; vz0=3; az0=0;
%q=[b1 b2 b3 alpha_c    alpha_d    beta_c    beta_d    phi_c phi_d px py pz ] vector of unknowns
q=[-2; -3; 4; deg2rad(47.467); deg2rad(-45.25); deg2rad(42.8); deg2rad(134.99); 0.52; 1; -2.1; -3.2; 4.1 ];

%solver initialization
counter=0;
for t=0:0.1:30
    counter=counter+1;
    res=10;
    iter=1;
    while res>0.00001 && iter<100
        Fq=[a1(1)+q(1)+e1(1)-q(10);
            a1(2)+c*cos(q(4))+d*cos(q(5))+e1(2)-q(11);
            a1(3)+c*sin(q(4))+d*sin(q(5))+e1(3)-q(12);
            a2(1)+c*sin(q(6))+d*sin(q(7))+e2(1)-q(10);
            a2(2)+q(2)+e2(2)-q(11);
            a2(3)+c*cos(q(6))+d*cos(q(7))+e2(3)-q(12);
            a3(1)+c*cos(q(8))+d*cos(q(9))+e3(1)-q(10);
            a3(2)+c*sin(q(8))+d*sin(q(9))+e3(2)-q(11);
            a3(3)+q(3)+e3(3)-q(12);
            q(10)-px0-vx0*t-0.5*ax0*t^2;
            q(11)-py0-vy0*t-0.5*ay0*t^2;
            q(12)-pz0-vz0*t-0.5*az0*t^2;
        ];

        J=[1 0 0 0      0      0      0      0      0      -1 0 0;
           0 0 0 -c*sin(q(4)) -d*sin(q(5)) 0      0      0      0      0 -1 0;
           0 0 0 c*cos(q(4)) d*cos(q(5)) 0      0      0      0      0 0 -1;
           0 0 0 0      c*cos(q(6)) d*cos(q(7)) 0      0      0      -1 0 0;
           0 1 0 0      0      0      0      0      0      0 -1 0;
           0 0 0 0      -c*sin(q(6)) -d*sin(q(7)) 0      0      0      0 -1;
           0 0 0 0      0      0      -c*sin(q(8)) -d*sin(q(9)) -1 0 0;
           0 0 0 0      0      0      c*cos(q(8)) d*cos(q(9)) 0 -1 0;
           0 0 1 0      0      0      0      0      0      0 0 -1;
           0 0 0 0      0      0      0      0      0      0 1 0;
           0 0 0 0      0      0      0      0      0      0 1 0;
           0 0 0 0      0      0      0      0      0      0 0 1;
        ];

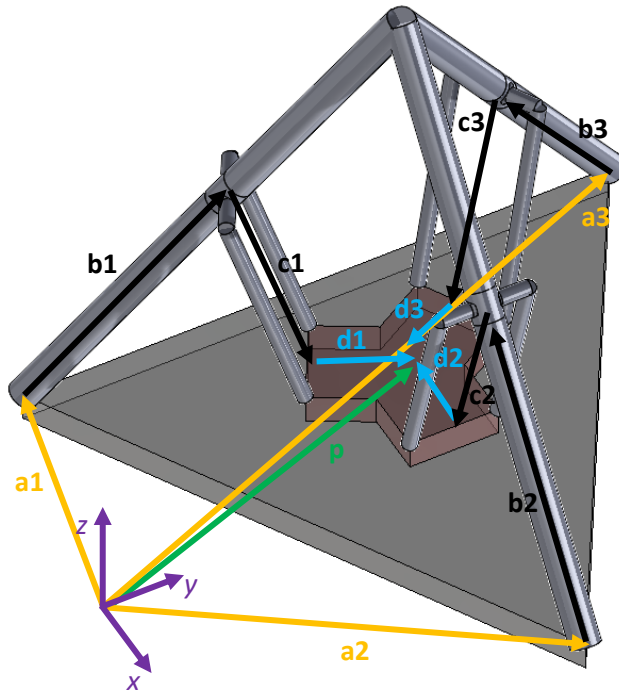
        qf = q - \J\Fq;
        q = qf;
        res = norm(Fq);
        iter=iter+1;
    end
    if iter>=100
        disp('no convergency, number of iterations exceeded')
    end
    P(:,counter)=qf;
    %velocity analysis
    vi=-J\J[0; 0; 0; 0; 0; 0; 0; 0; 0; -vx0-ax0*t; -vy0-ay0*t; -vz0-az0*t];
    V(:,counter)=vi;
    %acceleration analysis
    jp=[0 0 0 0      0      0      0      0      0      0 0 0;
        0 0 0 -c*vi(4)*cos(q(4))-d*vi(5)*cos(q(5)) 0      0      0      0      0      0 0 0;
        0 0 0 -c*vi(4)*sin(q(4))-d*vi(5)*sin(q(5)) 0      0      0      0      0      0 0 0;
        0 0 0 0      -c*vi(6)*sin(q(6))-d*vi(7)*sin(q(7)) 0      0      0      0 0 0;
        0 0 0 0      0      0      0      0      0      0 0 0;
        0 0 0 0      -c*vi(6)*cos(q(6))-d*vi(7)*cos(q(7)) 0      0      0      0 0 0;
        0 0 0 0      0      0      -c*vi(8)*cos(q(8))-d*vi(9)*cos(q(9)) 0 0 0;
        0 0 0 0      0      0      -c*vi(8)*sin(q(8))-d*vi(9)*sin(q(9)) 0 0 0;
        0 0 0 0      0      0      0      0      0      0 0 0;
        0 0 0 0      0      0      0      0      0      0 0 0;
        0 0 0 0      0      0      0      0      0      0 0 0;
        0 0 0 0      0      0      0      0      0      0 0 0;
    ];
    ai=-J\(-jp*vi-[0; 0; 0; 0; 0; 0; 0; 0; 0; -ax0*t; -az0*t]);
    A(:,counter)=ai;
end
```

```

%animate the position solution
nSteps=size(P,2);
figure
%pause
for i=1:nSteps
    O=[0; 0; 0];
    %first arm
    A=a1;
    B=A+[P(1,i);0;0];
    C=B+[0;c*cos(P(4,i));c*sin(P(4,i))];
    D=C+[0;d*cos(P(5,i));d*sin(P(5,i))];
    E=D+e1;
    cla
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)])
    hold on
    plot3([B(1) C(1)], [B(2) C(2)], [B(3) C(3)])
    plot3([C(1) D(1)], [C(2) D(2)], [C(3) D(3)])
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)])
    %second arm
    A=a2;
    B=A+[0;P(2,i);0];
    C=B+[c*cos(P(6,i));0;c*sin(P(6,i))];
    D=C+[d*cos(P(7,i));0;d*sin(P(7,i))];
    E=D+e2;
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)])
    hold on
    plot3([B(1) C(1)], [B(2) C(2)], [B(3) C(3)])
    plot3([C(1) D(1)], [C(2) D(2)], [C(3) D(3)])
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)])
    %third arm
    A=a3;
    B=A+[0;0;P(3,i)];
    C=B+[c*cos(P(8,i));c*sin(P(8,i));0];
    D=C+[d*cos(P(9,i));d*sin(P(9,i));0];
    E=D+e3;
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)])
    hold on
    plot3([B(1) C(1)], [B(2) C(2)], [B(3) C(3)])
    plot3([C(1) D(1)], [C(2) D(2)], [C(3) D(3)])
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)])
    xlabel('x')
    ylabel('y')
    zlabel('z')
    title('animation of the position')
    axis([-300 0 -300 0 0 300])
    view(129.6260,39.6168)
    pause(0.01)
    grid on
end

```

10. Linear delta



$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 + \mathbf{c}_1 + \mathbf{d}_1 - \mathbf{p} \\ \mathbf{a}_2 + \mathbf{b}_2 + \mathbf{c}_2 + \mathbf{d}_2 - \mathbf{p} \\ \mathbf{a}_3 + \mathbf{b}_3 + \mathbf{c}_3 + \mathbf{d}_3 - \mathbf{p} \end{bmatrix}$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_1 \cos(\alpha_{b1}) + c \cos(\alpha_{c1}) + d_{1x} - p_x \\ a_{1y} + b_1 \cos(\beta_{b1}) + c \cos(\beta_{c1}) + d_{1y} - p_y \\ a_{1z} + b_1 \cos(\gamma_{b1}) + c \cos(\gamma_{c1}) + d_{1z} - p_z \\ a_{2x} + b_2 \cos(\alpha_{b2}) + c \cos(\alpha_{c2}) + d_{2x} - p_x \\ a_{2y} + b_2 \cos(\beta_{b2}) + c \cos(\beta_{c2}) + d_{2y} - p_y \\ a_{2z} + b_2 \cos(\gamma_{b2}) + c \cos(\gamma_{c2}) + d_{2z} - p_z \\ a_{3x} + b_3 \cos(\alpha_{b3}) + c \cos(\alpha_{c3}) + d_{3x} - p_x \\ a_{3y} + b_3 \cos(\beta_{b3}) + c \cos(\beta_{c3}) + d_{3y} - p_y \\ a_{3z} + b_3 \cos(\gamma_{b3}) + c \cos(\gamma_{c3}) + d_{3z} - p_z \end{bmatrix}$$

$$\mathbf{q} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \alpha_{c1} \\ \beta_{c1} \\ \gamma_{c1} \\ \alpha_{c2} \\ \beta_{c2} \\ \gamma_{c2} \\ \alpha_{c3} \\ \beta_{c3} \\ \gamma_{c3} \end{bmatrix}$$

Notice that there are 12 unknowns and 9 constraint equations. This is because we are using 4 parameters to describe vector $\mathbf{c}$, the magnitude c and three angles (α, β, γ) to describe the orientation. But we need

to remember that a vector in space is described by 3 parameters, for instance the magnitude c and 2 angles (elevation and azimuth). For the system of equations this means that we either (i) re-express the 3 angles in terms of other 2 angles, or (ii) we add additional equations relating the three angles for each vector c . In this case we opt for the latter, add additional equations.

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_1 \cos(\alpha_{b1}) + c \cos(\alpha_{c1}) + d_{1x} - p_x \\ a_{1y} + b_1 \cos(\beta_{b1}) + c \cos(\beta_{c1}) + d_{1y} - p_y \\ a_{1z} + b_1 \cos(\gamma_{b1}) + c \cos(\gamma_{c1}) + d_{1z} - p_z \\ a_{2x} + b_2 \cos(\alpha_{b2}) + c \cos(\alpha_{c2}) + d_{2x} - p_x \\ a_{2y} + b_2 \cos(\beta_{b2}) + c \cos(\beta_{c2}) + d_{2y} - p_y \\ a_{2z} + b_2 \cos(\gamma_{b2}) + c \cos(\gamma_{c2}) + d_{2z} - p_z \\ a_{3x} + b_3 \cos(\alpha_{b3}) + c \cos(\alpha_{c3}) + d_{3x} - p_x \\ a_{3y} + b_3 \cos(\beta_{b3}) + c \cos(\beta_{c3}) + d_{3y} - p_y \\ a_{3z} + b_3 \cos(\gamma_{b3}) + c \cos(\gamma_{c3}) + d_{3z} - p_z \\ (\cos \alpha_{c1})^2 + (\cos \beta_{c1})^2 + (\cos \gamma_{c1})^2 - 1 \\ (\cos \alpha_{c2})^2 + (\cos \beta_{c2})^2 + (\cos \gamma_{c2})^2 - 1 \\ (\cos \alpha_{c3})^2 + (\cos \beta_{c3})^2 + (\cos \gamma_{c3})^2 - 1 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \alpha_{c1} \\ \beta_{c1} \\ \gamma_{c1} \\ \alpha_{c2} \\ \beta_{c2} \\ \gamma_{c2} \\ \alpha_{c3} \\ \beta_{c3} \\ \gamma_{c3} \end{bmatrix}$$

We add the driving constraints,

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_1 \cos(\alpha_{b1}) + c \cos(\alpha_{c1}) + d_{1x} - p_x \\ a_{1y} + b_1 \cos(\beta_{b1}) + c \cos(\beta_{c1}) + d_{1y} - p_y \\ a_{1z} + b_1 \cos(\gamma_{b1}) + c \cos(\gamma_{c1}) + d_{1z} - p_z \\ a_{2x} + b_2 \cos(\alpha_{b2}) + c \cos(\alpha_{c2}) + d_{2x} - p_x \\ a_{2y} + b_2 \cos(\beta_{b2}) + c \cos(\beta_{c2}) + d_{2y} - p_y \\ a_{2z} + b_2 \cos(\gamma_{b2}) + c \cos(\gamma_{c2}) + d_{2z} - p_z \\ a_{3x} + b_3 \cos(\alpha_{b3}) + c \cos(\alpha_{c3}) + d_{3x} - p_x \\ a_{3y} + b_3 \cos(\beta_{b3}) + c \cos(\beta_{c3}) + d_{3y} - p_y \\ a_{3z} + b_3 \cos(\gamma_{b3}) + c \cos(\gamma_{c3}) + d_{3z} - p_z \\ (\cos \alpha_{c1})^2 + (\cos \beta_{c1})^2 + (\cos \gamma_{c1})^2 - 1 \\ (\cos \alpha_{c2})^2 + (\cos \beta_{c2})^2 + (\cos \gamma_{c2})^2 - 1 \\ (\cos \alpha_{c3})^2 + (\cos \beta_{c3})^2 + (\cos \gamma_{c3})^2 - 1 \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \alpha_{c1} \\ \beta_{c1} \\ \gamma_{c1} \\ \alpha_{c2} \\ \beta_{c2} \\ \gamma_{c2} \\ \alpha_{c3} \\ \beta_{c3} \\ \gamma_{c3} \\ p_x \\ p_y \\ p_z \end{bmatrix}$$

Notice that the terms $\cos(\alpha_b)$, $\cos(\beta_b)$, $\cos(\gamma_b)$ (second term on row 1 to 9), are the cartesian components of the unitary vectors pointing in the direction of the sliders. These components are constant, hence vanish in the time derivative of the Jacobian matrix ($\mathbf{J}$ dot).

$$\mathbf{J} = \begin{bmatrix} \cos(\alpha_{b1}) & 0 & 0 & -\sin(\alpha_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ \cos(\beta_{b1}) & 0 & 0 & 0 & -\sin(\beta_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ \cos(\gamma_{b1}) & 0 & 0 & 0 & 0 & -\sin(\gamma_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & \cos(\alpha_{b2}) & 0 & 0 & 0 & 0 & -\sin(\alpha_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & \cos(\beta_{b2}) & 0 & 0 & 0 & 0 & 0 & -\sin(\beta_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & \cos(\gamma_{b2}) & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\gamma_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & \cos(\alpha_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\alpha_{c3}) & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & \cos(\beta_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\beta_{c3}) & 0 & 0 & -1 & 0 \\ 0 & 0 & \cos(\gamma_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\gamma_{c3}) & 0 & 0 & -1 \\ 0 & 0 & 0 & -2\cos\alpha_{c1}\sin\alpha_{c1} & -2\cos\beta_{c1}\sin\beta_{c1} & -2\cos\gamma_{c1}\sin\gamma_{c1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2\cos\alpha_{c2}\sin\alpha_{c2} & -2\cos\beta_{c2}\sin\beta_{c2} & -2\cos\gamma_{c2}\sin\gamma_{c2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\cos\alpha_{c3}\sin\alpha_{c3} & -2\cos\beta_{c3}\sin\beta_{c3} & -2\cos\gamma_{c3}\sin\gamma_{c3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Notice that $-2\cos(x)\sin(x)$ is equal to $-\sin(2x)$

$$\mathbf{J} = \begin{bmatrix} \cos(\alpha_{b1}) & 0 & 0 & -\sin(\alpha_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ \cos(\beta_{b1}) & 0 & 0 & 0 & -\sin(\beta_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ \cos(\gamma_{b1}) & 0 & 0 & 0 & 0 & -\sin(\gamma_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & \cos(\alpha_{b2}) & 0 & 0 & 0 & 0 & -\sin(\alpha_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & \cos(\beta_{b2}) & 0 & 0 & 0 & 0 & 0 & -\sin(\beta_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & \cos(\gamma_{b2}) & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\gamma_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & \cos(\alpha_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\alpha_{c3}) & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & \cos(\beta_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\beta_{c3}) & 0 & 0 & -1 & 0 \\ 0 & 0 & \cos(\gamma_{b3}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(\gamma_{c3}) & 0 & 0 & -1 \\ 0 & 0 & 0 & -\sin(2\alpha_{c1}) & -\sin(2\beta_{c1}) & -\sin(2\gamma_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\sin(2\alpha_{c2}) & -\sin(2\beta_{c2}) & -\sin(2\gamma_{c2}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\sin(2\alpha_{c3}) & -\sin(2\beta_{c3}) & -\sin(2\gamma_{c3}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\dot{\mathbf{J}} = \begin{bmatrix} 0 & 0 & 0 & -c \cos(\alpha_{c1}) \dot{\alpha}_{c1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -c \cos(\beta_{c1}) \dot{\beta}_{c1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -c \cos(\gamma_{c1}) \dot{\gamma}_{c1} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\alpha_{c2}) \dot{\alpha}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\beta_{c2}) \dot{\beta}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\gamma_{c2}) \dot{\gamma}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\alpha_{c3}) \dot{\alpha}_{c3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\beta_{c3}) \dot{\beta}_{c3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\gamma_{c3}) \dot{\gamma}_{c3} & 0 & 0 & 0 \\ 0 & 0 & 0 & -2\dot{\alpha}_{c1} \cos(2\alpha_{c1}) & -2\dot{\beta}_{c1} \cos(2\beta_{c1}) & -2\dot{\gamma}_{c1} \cos(2\gamma_{c1}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -2\dot{\alpha}_{c2} \cos(2\alpha_{c2}) & -2\dot{\beta}_{c2} \cos(2\beta_{c2}) & -2\dot{\gamma}_{c2} \cos(2\gamma_{c2}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -2\dot{\alpha}_{c3} \cos(2\alpha_{c3}) & -2\dot{\beta}_{c3} \cos(2\beta_{c3}) & -2\dot{\gamma}_{c3} \cos(2\gamma_{c3}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} \cos(\alpha_{b1}) & 0 & 0 & -c\sin(\alpha_{c1}) & 0 & 0 & 0 \\ \cos(\beta_{b1}) & 0 & 0 & 0 & -c\sin(\beta_{c1}) & 0 & 0 \\ \cos(\gamma_{b1}) & 0 & 0 & 0 & 0 & -c\sin(\gamma_{c1}) & 0 \\ 0 & \cos(\alpha_{b2}) & 0 & 0 & 0 & 0 & -c\sin(\alpha_{c2}) \\ 0 & \cos(\beta_{b2}) & 0 & 0 & 0 & 0 & 0 \\ 0 & \cos(\gamma_{b2}) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos(\alpha_{b3}) & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos(\beta_{b3}) & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos(\gamma_{b3}) & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2\alpha_{c1} & 2\beta_{c1} & 2\gamma_{c1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2\alpha_{c2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ -c\sin(\beta_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & -c\sin(\gamma_{c2}) & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & -c\sin(\alpha_{c3}) & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -c\sin(\beta_{c3}) & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -c\sin(\gamma_{c3}) & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2\beta_{c2} & 2\gamma_{c2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2\alpha_{c3} & 2\beta_{c3} & 2\gamma_{c3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{j} = \begin{bmatrix} 0 & 0 & 0 & -c \cos(\alpha_{c1}) \dot{\alpha}_{c1} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -c \cos(\beta_{c1}) \dot{\beta}_{c1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -c \cos(\gamma_{c1}) \dot{\gamma}_{c1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -c \cos(\alpha_{c2}) \dot{\alpha}_{c2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 \dot{\alpha}_{c1} & 2 \dot{\beta}_{c1} & 2 \dot{\gamma}_{c1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 \dot{\alpha}_{c2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -c \cos(\beta_{c2}) \dot{\beta}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -c \cos(\gamma_{c2}) \dot{\gamma}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -c \cos(\alpha_{c3}) \dot{\alpha}_{c3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c \cos(\beta_{c3}) \dot{\beta}_{c3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -c \cos(\gamma_{c3}) \dot{\gamma}_{c3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 \dot{\beta}_{c2} & 2 \dot{\gamma}_{c2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 \dot{\alpha}_{c3} & 2 \dot{\beta}_{c3} & 2 \dot{\gamma}_{c3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{\dot{q}} = \begin{bmatrix} \dot{b}_1 \\ \dot{b}_2 \\ \dot{b}_3 \\ \dot{\alpha}_{c1} \\ \dot{\beta}_{c1} \\ \dot{\gamma}_{c1} \\ \dot{\alpha}_{c2} \\ \dot{\beta}_{c2} \\ \dot{\gamma}_{c2} \\ \dot{\alpha}_{c3} \\ \dot{\beta}_{c3} \\ \dot{\gamma}_{c3} \\ \dot{p}_x \\ \dot{p}_y \\ \dot{p}_z \end{bmatrix} \quad \mathbf{\ddot{q}} = \begin{bmatrix} \ddot{b}_1 \\ \ddot{b}_2 \\ \ddot{b}_3 \\ \ddot{\alpha}_{c1} \\ \ddot{\beta}_{c1} \\ \ddot{\gamma}_{c1} \\ \ddot{\alpha}_{c2} \\ \ddot{\beta}_{c2} \\ \ddot{\gamma}_{c2} \\ \ddot{\alpha}_{c3} \\ \ddot{\beta}_{c3} \\ \ddot{\gamma}_{c3} \\ \ddot{p}_x \\ \ddot{p}_y \\ \ddot{p}_z \end{bmatrix} \quad \dot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\dot{p}_x^0 - \ddot{p}_x^0 t \\ -\dot{p}_y^0 - \ddot{p}_y^0 t \\ -\dot{p}_z^0 - \ddot{p}_z^0 t \end{bmatrix} \quad \ddot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\ddot{p}_x^0 \\ -\ddot{p}_y^0 \\ -\ddot{p}_z^0 \end{bmatrix}$$

Matlab code

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% This code solves the kinematic analysis of the linear delta robot using
% multibody kinematics based on the use of close loop equation
% a: location of the origin for the linear actuators
% b: stroke of the linear actuators
% c: length of the links of the parallelogram
% d: location of the Poi with respect the middle point of the
% parallelogram's connecting joints in the effector
% q: vector with the kinematic variables
% P: Matrix with the historic position
% V: Matrix with the historic velocities
% A: Matrix with the historic acceleration
% Fq: Vector with kinematic constraints
% I: Jacobian matrix
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all
clc
%close all
% the GCS is located at the center of the robot base, center of the triangle
%vector from gcs to origin of sliders
a1=[230*cosd(-120); 230*sind(-120); 0];
a2=[230; 0; 0];
a3=[230*cosd(120); 230*sind(120); 0];
% unit vector along the prismatic joints
ub1=[cosd(73.29);cosd(60.13);cosd(35.1)];
ub2=[cosd(125.1);cosd(90);cosd(35.1)];
ub3=[cosd(73.29);cosd(119.87);cosd(35.1)];
%length of parallelogram links
c=150;
%distance of the effector Poi to the effector joints
d1=[60*cosd(60); 60*sind(60); 0];
d2=[-60; 0; 0];
d3=[60*cosd(-60); 60*sind(-60); 0];
%driving constraints
px0=0; vx0=30; ax0=0;
py0=0; vy0=30; ay0=0;
pz0=25; vz0=0; az0=0;
%vector of unknowns
%q=[b1 b2 b3 alpha_c1 beta_c1 gamma_c1 alpha_c2 beta_c2 gamma_c2 alpha_c3 beta_c3 gamma_c3 px py pz ]
q=[153; 153; 153; deg2rad(74.467); deg2rad(61.25); deg2rad(213.8); deg2rad(122.99); deg2rad(89); deg2rad(147.99); deg2rad(74); deg2rad(116); deg2rad(148); 2.1; 3.2; 4.1];
%this initial guess for q is for the solution of effector at (0,0,0)
%solver initialization
counter=0;
for t=0:0.1:30
    counter=counter+1;
    res=10;
    iter=1;
    while res>0.00001 && iter<100
        Fq=[a1(1)+q(1)*ub1(1)+c*cos(q(4))+d1(1)-q(13);
            a1(2)+q(1)*ub1(2)+c*cos(q(5))+d1(2)-q(14);
            a1(3)+q(1)*ub1(3)+c*cos(q(6))+d1(3)-q(15);
            a2(1)+q(2)*ub2(1)+c*cos(q(7))+d2(1)-q(13);
            a2(2)+q(2)*ub2(2)+c*cos(q(8))+d2(2)-q(14);
            a2(3)+q(2)*ub2(3)+c*cos(q(9))+d2(3)-q(15);
            a3(1)+q(3)*ub3(1)+c*cos(q(10))+d3(1)-q(13);
            a3(2)+q(3)*ub3(2)+c*cos(q(11))+d3(2)-q(14);
            a3(3)+q(3)*ub3(3)+c*cos(q(12))+d3(3)-q(15);
            cos(q(4))^2+cos(q(5))^2+cos(q(6))^2-1
            cos(q(7))^2+cos(q(8))^2+cos(q(9))^2-1
            cos(q(10))^2+cos(q(11))^2+cos(q(12))^2-1
            q(13)-px0-vx0*cos(0.4*t)-0.5*ax0*t^2;
            q(14)-py0-vy0*sin(0.4*t)-0.5*ay0*t^2;
            q(15)-pz0-vz0*t-0.5*az0*t^2;
            ];

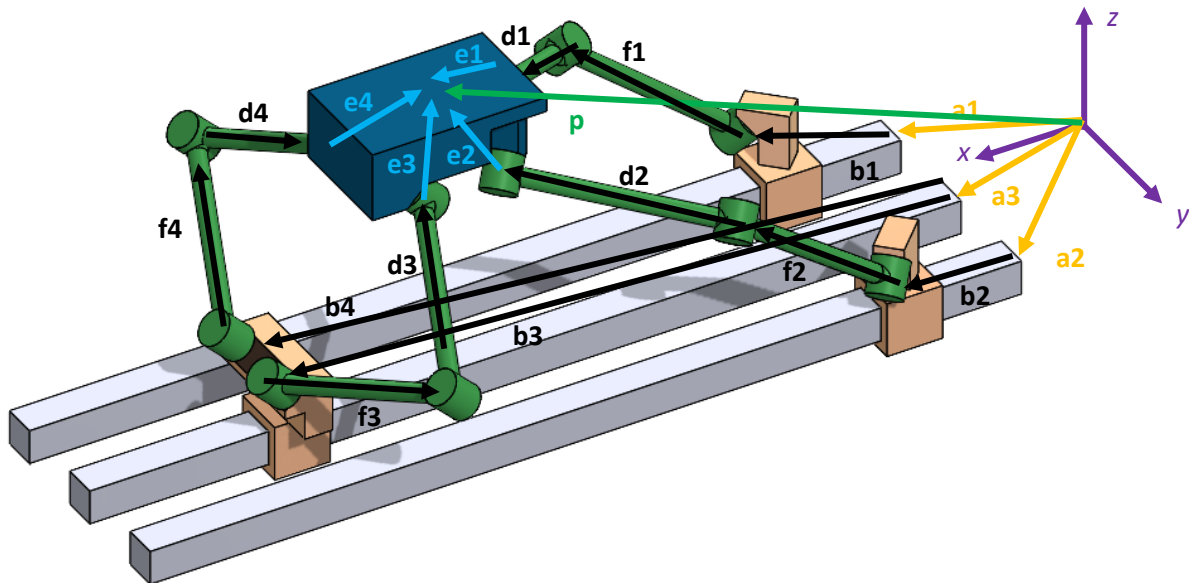
        J=[ub1(1),0, 0, -c*sin(q(4)), 0, 0, 0, 0, 0, 0, 0, 0, -1, 0, 0;
            ub1(2),0, 0, 0, -c*sin(q(5)), 0, 0, 0, 0, 0, 0, 0, 0, -1, 0;
            ub1(3),0, 0, 0, 0, -c*sin(q(6)), 0, 0, 0, 0, 0, 0, 0, 0, -1;
            0, ub2(1),0, 0, 0, 0, 0, -c*sin(q(7)), 0, 0, 0, 0, 0, -1, 0;
            0, ub2(2),0, 0, 0, 0, 0, 0, -c*sin(q(8)), 0, 0, 0, 0, 0, -1;
            0, ub2(3),0, 0, 0, 0, 0, 0, 0, -c*sin(q(9)), 0, 0, 0, 0, -1;
            0, 0, ub3(1),0, 0, 0, 0, 0, 0, 0, -c*sin(q(10)), 0, 0, -1, 0;
            0, 0, ub3(2),0, 0, 0, 0, 0, 0, 0, 0, -c*sin(q(11)), 0, 0, -1;
            0, 0, ub3(3),0, 0, 0, 0, 0, 0, 0, 0, 0, -c*sin(q(12)),0, 0, -1;
            0, 0, 0, -sin(2*q(4)), -sin(2*q(5)), -sin(2*q(6)), 0, 0, 0, 0, 0, 0, 0, 0, 0;
            0, 0, 0, 0, 0, -sin(2*q(7)), -sin(2*q(8)), -sin(2*q(9)), 0, 0, 0, 0, 0, 0;
            0, 0, 0, 0, 0, 0, 0, 0, -sin(2*q(10)), -sin(2*q(11)), -sin(2*q(12)),0, 0, 0;
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0;
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0;
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1;
            ];

        qf = q - \Fq;
        q = qf;
        res = norm(Fq);
        iter=iter+1;
    end
    if iter>=100
        disp(strcat('no convergency, number of iterations exceeded. frame=',num2str(counter)))
        break
    end
    P(:,counter)=qf;
end

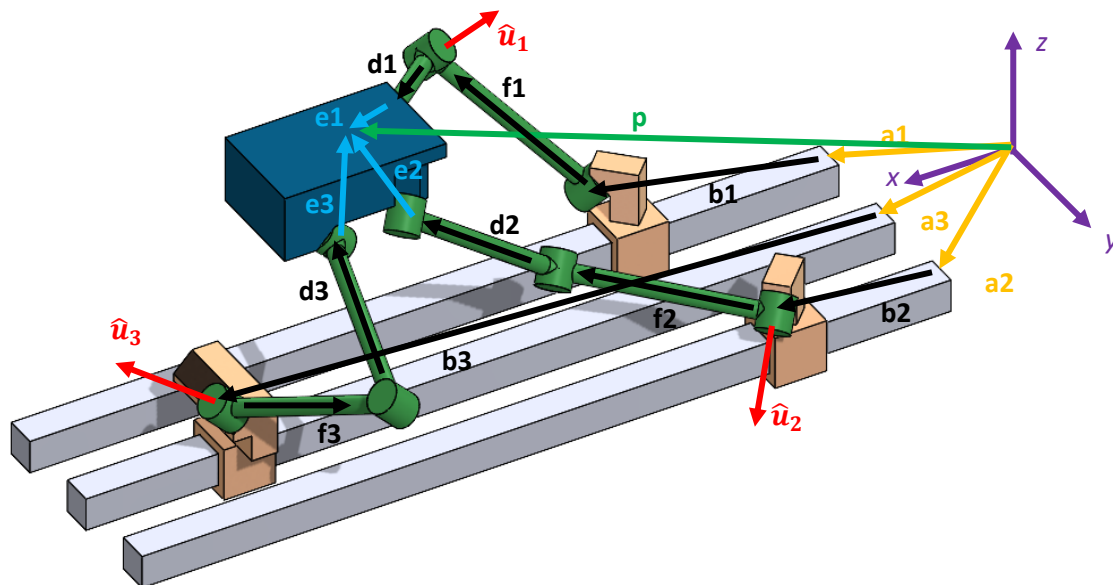
```

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11. Linear Tripteron – Parallel and coplanar sliders



Notice that leg 4 is redundant, meaning that this leg is not necessary for the constraining of the effector to exhibit 3-DoF. This leg is added to provide a better distribution of the loads. So, this leg should not be included into the system of close loop equations.



The three close-loop equations are as follows:

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 + \mathbf{f}_1 + \mathbf{d}_1 + \mathbf{e}_1 - \mathbf{p} \\ \mathbf{a}_2 + \mathbf{b}_2 + \mathbf{f}_2 + \mathbf{d}_2 + \mathbf{e}_2 - \mathbf{p} \\ \mathbf{a}_3 + \mathbf{b}_3 + \mathbf{f}_3 + \mathbf{d}_3 + \mathbf{e}_3 - \mathbf{p} \end{bmatrix}$$

Expanding the equations into its three components we have:

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1 \cos(\alpha_{f1}) + d_1 \cos(\alpha_{d1}) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1 \cos(\beta_{f1}) + d_1 \cos(\beta_{d1}) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1 \cos(\gamma_{f1}) + d_1 \cos(\gamma_{d1}) + e_{1z} - p_z \\ a_{2x} + b_{2x} + f_2 \cos(\alpha_{f2}) + d_2 \cos(\alpha_{d2}) + e_{2x} - p_x \\ a_{2y} + b_{2y} + f_2 \cos(\beta_{f2}) + d_2 \cos(\beta_{d2}) + e_{2y} - p_y \\ a_{2z} + b_{2z} + f_2 \cos(\gamma_{f2}) + d_2 \cos(\gamma_{d2}) + e_{2z} - p_z \\ a_{3x} + b_{3x} + f_3 \cos(\alpha_{f3}) + d_3 \cos(\alpha_{d3}) + e_{3x} - p_x \\ a_{3y} + b_{3y} + f_3 \cos(\beta_{f3}) + d_3 \cos(\beta_{d3}) + e_{3y} - p_y \\ a_{3z} + b_{3z} + f_3 \cos(\gamma_{f3}) + d_3 \cos(\gamma_{d3}) + e_{3z} - p_z \end{bmatrix}_{9 \times 1}$$

The vector of unknowns is as follows:

$$\mathbf{q} = [b_{1x} \ b_{2x} \ b_{3x} \ \alpha_{f1} \ \beta_{f1} \ \gamma_{f1} \ \alpha_{f2} \ \beta_{f2} \ \gamma_{f2} \ \alpha_{f3} \ \beta_{f3} \ \gamma_{f3} \ \alpha_{d1} \ \beta_{d1} \ \gamma_{d1} \ \alpha_{d2} \ \beta_{d2} \ \gamma_{d2} \ \alpha_{d3} \ \beta_{d3} \ \gamma_{d3}]^T_{21 \times 1}$$

Observe that the number of unknowns is not equal to the number of equations. To understand this we need to remember that a vector in space is described by only 3 parameters. Since we are using 4 parameters to describe vectors $\mathbf{f}_i$ and $\mathbf{d}_i$ (magnitude and three angles α, β, γ to describe the orientation) we need to add a constraint equation to each of the six vectors to account for the extra parameter.

Additionally, notice that all the 6 vectors are constrained to rotate inside a plane. This means that we need to add an additional constraint to each vector in the form of an inner product of the vectors with respect to unit vectors which are normal to the plane in which the vector rotates.

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1 \cos(\alpha_{f1}) + d_1 \cos(\alpha_{d1}) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1 \cos(\beta_{f1}) + d_1 \cos(\beta_{d1}) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1 \cos(\gamma_{f1}) + d_1 \cos(\gamma_{d1}) + e_{1z} - p_z \\ a_{2x} + b_{2x} + f_2 \cos(\alpha_{f2}) + d_2 \cos(\alpha_{d2}) + e_{2x} - p_x \\ a_{2y} + b_{2y} + f_2 \cos(\beta_{f2}) + d_2 \cos(\beta_{d2}) + e_{2y} - p_y \\ a_{2z} + b_{2z} + f_2 \cos(\gamma_{f2}) + d_2 \cos(\gamma_{d2}) + e_{2z} - p_z \\ a_{3x} + b_{3x} + f_3 \cos(\alpha_{f3}) + d_3 \cos(\alpha_{d3}) + e_{3x} - p_x \\ a_{3y} + b_{3y} + f_3 \cos(\beta_{f3}) + d_3 \cos(\beta_{d3}) + e_{3y} - p_y \\ a_{3z} + b_{3z} + f_3 \cos(\gamma_{f3}) + d_3 \cos(\gamma_{d3}) + e_{3z} - p_z \\ \cos^2(\alpha_{f1}) + \cos^2(\beta_{f1}) + \cos^2(\gamma_{f1}) - 1 \\ \cos^2(\alpha_{f2}) + \cos^2(\beta_{f2}) + \cos^2(\gamma_{f2}) - 1 \\ \cos^2(\alpha_{f3}) + \cos^2(\beta_{f3}) + \cos^2(\gamma_{f3}) - 1 \\ \cos^2(\alpha_{d1}) + \cos^2(\beta_{d1}) + \cos^2(\gamma_{d1}) - 1 \\ \cos^2(\alpha_{d2}) + \cos^2(\beta_{d2}) + \cos^2(\gamma_{d2}) - 1 \\ \cos^2(\alpha_{d3}) + \cos^2(\beta_{d3}) + \cos^2(\gamma_{d3}) - 1 \\ u_{1x} \cos(\alpha_{f1}) + u_{1y} \cos(\beta_{f1}) + u_{1z} \cos(\gamma_{f1}) \\ u_{1x} \cos(\alpha_{d1}) + u_{1y} \cos(\beta_{d1}) + u_{1z} \cos(\gamma_{d1}) \\ u_{2x} \cos(\alpha_{f2}) + u_{2y} \cos(\beta_{f2}) + u_{2z} \cos(\gamma_{f2}) \\ u_{2x} \cos(\alpha_{d2}) + u_{2y} \cos(\beta_{d2}) + u_{2z} \cos(\gamma_{d2}) \\ u_{3x} \cos(\alpha_{f3}) + u_{3y} \cos(\beta_{f3}) + u_{3z} \cos(\gamma_{f3}) \\ u_{3x} \cos(\alpha_{d3}) + u_{3y} \cos(\beta_{d3}) + u_{3z} \cos(\gamma_{d3}) \end{bmatrix}_{21 \times 1}$$

Finally, we can add the driving constraints:

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1 \cos(\alpha_{f1}) + d_1 \cos(\alpha_{d1}) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1 \cos(\beta_{f1}) + d_1 \cos(\beta_{d1}) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1 \cos(\gamma_{f1}) + d_1 \cos(\gamma_{d1}) + e_{1z} - p_z \\ a_{2x} + b_{2x} + f_2 \cos(\alpha_{f2}) + d_2 \cos(\alpha_{d2}) + e_{2x} - p_x \\ a_{2y} + b_{2y} + f_2 \cos(\beta_{f2}) + d_2 \cos(\beta_{d2}) + e_{2y} - p_y \\ a_{2z} + b_{2z} + f_2 \cos(\gamma_{f2}) + d_2 \cos(\gamma_{d2}) + e_{2z} - p_z \\ a_{3x} + b_{3x} + f_3 \cos(\alpha_{f3}) + d_3 \cos(\alpha_{d3}) + e_{3x} - p_x \\ a_{3y} + b_{3y} + f_3 \cos(\beta_{f3}) + d_3 \cos(\beta_{d3}) + e_{3y} - p_y \\ a_{3z} + b_{3z} + f_3 \cos(\gamma_{f3}) + d_3 \cos(\gamma_{d3}) + e_{3z} - p_z \\ \cos^2(\alpha_{f1}) + \cos^2(\beta_{f1}) + \cos^2(\gamma_{f1}) - 1 \\ \cos^2(\alpha_{f2}) + \cos^2(\beta_{f2}) + \cos^2(\gamma_{f2}) - 1 \\ \cos^2(\alpha_{f3}) + \cos^2(\beta_{f3}) + \cos^2(\gamma_{f3}) - 1 \\ \cos^2(\alpha_{d1}) + \cos^2(\beta_{d1}) + \cos^2(\gamma_{d1}) - 1 \\ \cos^2(\alpha_{d2}) + \cos^2(\beta_{d2}) + \cos^2(\gamma_{d2}) - 1 \\ \cos^2(\alpha_{d3}) + \cos^2(\beta_{d3}) + \cos^2(\gamma_{d3}) - 1 \\ u_{1x} \cos(\alpha_{f1}) + u_{1y} \cos(\beta_{f1}) + u_{1z} \cos(\gamma_{f1}) \\ u_{1x} \cos(\alpha_{d1}) + u_{1y} \cos(\beta_{d1}) + u_{1z} \cos(\gamma_{d1}) \\ u_{2x} \cos(\alpha_{f2}) + u_{2y} \cos(\beta_{f2}) + u_{2z} \cos(\gamma_{f2}) \\ u_{2x} \cos(\alpha_{d2}) + u_{2y} \cos(\beta_{d2}) + u_{2z} \cos(\gamma_{d2}) \\ u_{3x} \cos(\alpha_{f3}) + u_{3y} \cos(\beta_{f3}) + u_{3z} \cos(\gamma_{f3}) \\ u_{3x} \cos(\alpha_{d3}) + u_{3y} \cos(\beta_{d3}) + u_{3z} \cos(\gamma_{d3}) \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix}_{24 \times 1}$$

$$\mathbf{q} = \begin{bmatrix} b_{1x} \\ b_{2x} \\ b_{3x} \\ \alpha_{f1} \\ \beta_{f1} \\ \gamma_{f1} \\ \alpha_{f2} \\ \beta_{f2} \\ \gamma_{f2} \\ \alpha_{f3} \\ \beta_{f3} \\ \gamma_{f3} \\ \alpha_{d1} \\ \beta_{d1} \\ \gamma_{d1} \\ \alpha_{d2} \\ \beta_{d2} \\ \gamma_{d2} \\ \alpha_{d3} \\ \beta_{d3} \\ \gamma_{d3} \\ p_x \\ p_y \\ p_z \end{bmatrix}_{24 \times 1}$$

To solve this whole system of equations we need to compute the Jacobian matrix, which for this problem is a matrix of 24 by 24. Instead of solving this problem we will reduce the complexity of the problem by setting the orientation of each link in terms of only one variable rotation angle and two constant angles. This approach will lead to a system of equations of size 12 by 1, with a Jacobian matrix with size of 12 by 12.

Here is important to remark that each close loop equation, which is modeling each arm of the robot, can be solved independently from the other loop equations. For instance, the first arm of the robot can be solved as follows:

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1 \cos(\alpha_{f1}) + d_1 \cos(\alpha_{d1}) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1 \cos(\beta_{f1}) + d_1 \cos(\beta_{d1}) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1 \cos(\gamma_{f1}) + d_1 \cos(\gamma_{d1}) + e_{1z} - p_z \\ \cos^2(\alpha_{f1}) + \cos^2(\beta_{f1}) + \cos^2(\gamma_{f1}) - 1 \\ \cos^2(\alpha_{d1}) + \cos^2(\beta_{d1}) + \cos^2(\gamma_{d1}) - 1 \\ u_{1x} \cos(\alpha_{f1}) + u_{1y} \cos(\beta_{f1}) + u_{1z} \cos(\gamma_{f1}) \\ u_{1x} \cos(\alpha_{d1}) + u_{1y} \cos(\beta_{d1}) + u_{1z} \cos(\gamma_{d1}) \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix}_{10 \times 1} \quad \mathbf{q} = \begin{bmatrix} b_{1x} \\ \alpha_{f1} \\ \beta_{f1} \\ \gamma_{f1} \\ \alpha_{d1} \\ \beta_{d1} \\ \gamma_{d1} \\ p_x \\ p_y \\ p_z \end{bmatrix}_{10 \times 1}$$

This system of equations is solved three times, one for each arm of the robot.

Simplification of the system of equations

Here we will try to simplify the constraints equations by expressing the rotation of the links in terms of just one parameter.

We make use of the rotation matrix defined in terms of the Tait-Bryan angles with the sequence ZYX which is the one that accommodates to reference system presented in previous image.

$$z_{\alpha} y_{\beta} x_{\gamma} = \mathbf{A} = \begin{bmatrix} \cos(\alpha)\cos(\beta) & \cos(\alpha)\sin(\beta)\sin(\gamma) - \cos(\gamma)\sin(\alpha) & \sin(\alpha)\sin(\gamma) + \cos(\alpha)\cos(\gamma)\sin(\beta) \\ \cos(\beta)\sin(\alpha) & \cos(\alpha)\cos(\gamma) + \sin(\alpha)\sin(\beta)\sin(\gamma) & \cos(\gamma)\sin(\alpha)\sin(\beta) - \cos(\alpha)\sin(\gamma) \\ -\sin(\beta) & \cos(\beta)\sin(\gamma) & \cos(\beta)\cos(\gamma) \end{bmatrix}$$

We set a local coordinate system that initially coincides with the global coordinate system. On this local coordinate system, we define a vector $\mathbf{x}'$ that represents a link of the robot as follows:

$$\mathbf{x}' = \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}$$

By using this vector, we have that the constant rotations α and β define the orientation of the plane in which links $\mathbf{f}_i$ and $\mathbf{d}_i$ rotate. The rotation of the links is given by the γ angle.

Keep in mind that local vector $\mathbf{u} = [1 \ 0 \ 0]^T$ is useful for the definition of the unit vector $\hat{\mathbf{u}}_i$, and not should be confused with the local vector $\mathbf{r}$ used to define links $\mathbf{f}_i$.

$$\mathbf{r} = \mathbf{A}\mathbf{r} = \begin{bmatrix} \cos(\alpha)\cos(\beta) & \cos(\alpha)\sin(\beta)\sin(\gamma) - \cos(\gamma)\sin(\alpha) & \sin(\alpha)\sin(\gamma) + \cos(\alpha)\cos(\gamma)\sin(\beta) \\ \cos(\beta)\sin(\alpha) & \cos(\alpha)\cos(\gamma) + \sin(\alpha)\sin(\beta)\sin(\gamma) & \cos(\gamma)\sin(\alpha)\sin(\beta) - \cos(\alpha)\sin(\gamma) \\ -\sin(\beta) & \cos(\beta)\sin(\gamma) & \cos(\beta)\cos(\gamma) \end{bmatrix} \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix}$$

$$\mathbf{r} = \mathbf{A}\mathbf{r} = \begin{bmatrix} r(\cos(\alpha)\sin(\beta)\sin(\gamma) - \cos(\gamma)\sin(\alpha)) \\ r(\cos(\alpha)\cos(\gamma) + \sin(\alpha)\sin(\beta)\sin(\gamma)) \\ r\cos(\beta)\sin(\gamma) \end{bmatrix}$$

For links $\mathbf{f}_3$ and $\mathbf{d}_3$ we have that the rotation about the z axis is $\alpha_3=0$, and rotation about y axis is a constant value β :

$$\mathbf{A} = z_\alpha y_\beta x_\gamma = \begin{bmatrix} \cos(\beta) & \sin(\beta)\sin(\gamma) & \cos(\gamma)\sin(\beta) \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ -\sin(\beta) & \cos(\beta)\sin(\gamma) & \cos(\beta)\cos(\gamma) \end{bmatrix}$$

$$\mathbf{r} = \mathbf{A}\mathbf{r} = \begin{bmatrix} \cos(\beta) & \sin(\beta)\sin(\gamma) & \cos(\gamma)\sin(\beta) \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ -\sin(\beta) & \cos(\beta)\sin(\gamma) & \cos(\beta)\cos(\gamma) \end{bmatrix} \begin{bmatrix} 0 \\ r \\ 0 \end{bmatrix} = \begin{bmatrix} r\sin(\beta)\sin(\gamma) \\ r\cos(\gamma) \\ r\cos(\beta)\sin(\gamma) \end{bmatrix}$$

Now we can rewrite the close loop equation to set the kinematic constraints.

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 + \mathbf{f}_1 + \mathbf{d}_1 + \mathbf{e}_1 - \mathbf{p} \\ \mathbf{a}_2 + \mathbf{b}_2 + \mathbf{f}_2 + \mathbf{d}_2 + \mathbf{e}_2 - \mathbf{p} \\ \mathbf{a}_3 + \mathbf{b}_3 + \mathbf{f}_3 + \mathbf{d}_3 + \mathbf{e}_3 - \mathbf{p} \end{bmatrix}$$

Expanding the equations into its three components we have:

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1(\cos(\alpha_1)\sin(\beta_1)\sin(\gamma_{f1}) - \cos(\gamma_{f1})\sin(\alpha_1)) + d_1(\cos(\alpha_1)\sin(\beta_1)\sin(\gamma_{d1}) - \cos(\gamma_{d1})\sin(\alpha_1)) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1(\cos(\alpha_1)\cos(\gamma_{f1}) + \sin(\alpha_1)\sin(\beta_1)\sin(\gamma_{f1})) + d_1(\cos(\alpha_1)\cos(\gamma_{d1}) + \sin(\alpha_1)\sin(\beta_1)\sin(\gamma_{d1})) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1\cos(\beta_1)\sin(\gamma_{f1}) + d_1\cos(\beta_1)\sin(\gamma_{d1}) + e_{1z} - p_z \\ a_{2x} + b_{2x} + f_2(\cos(\alpha_2)\sin(\beta_2)\sin(\gamma_{f2}) - \cos(\gamma_{f2})\sin(\alpha_2)) + d_2(\cos(\alpha_2)\sin(\beta_2)\sin(\gamma_{d2}) - \cos(\gamma_{d2})\sin(\alpha_2)) + e_{2x} - p_x \\ a_{2y} + b_{2y} + f_2(\cos(\alpha_2)\cos(\gamma_{f2}) + \sin(\alpha_2)\sin(\beta_2)\sin(\gamma_{f2})) + d_2(\cos(\alpha_2)\cos(\gamma_{d2}) + \sin(\alpha_2)\sin(\beta_2)\sin(\gamma_{d2})) + e_{2y} - p_y \\ a_{2z} + b_{2z} + f_2\cos(\beta_2)\sin(\gamma_{f2}) + d_2\cos(\beta_2)\sin(\gamma_{d2}) + e_{2z} - p_z \\ a_{3x} + b_{3x} + f_3\sin(\beta_3)\sin(\gamma_{f3}) + d_3\sin(\beta_3)\sin(\gamma_{d3}) + e_{3x} - p_x \\ a_{3y} + b_{3y} + f_3\cos(\gamma_{f3}) + d_3\cos(\gamma_{d3}) + e_{3y} - p_y \\ a_{3z} + b_{3z} + f_3\cos(\beta_3)\sin(\gamma_{f3}) + d_3\cos(\beta_3)\sin(\gamma_{d3}) + e_{3z} - p_z \end{bmatrix}_{9 \times 1}$$

We add the driving constraints to the set of kinematic constraints.

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + b_{1x} + f_1 (\cos(\alpha_1) \sin(\beta_1) \sin(\gamma_{f1}) - \cos(\gamma_{f1}) \sin(\alpha_1)) + d_1 (\cos(\alpha_1) \sin(\beta_1) \sin(\gamma_{d1}) - \cos(\gamma_{d1}) \sin(\alpha_1)) + e_{1x} - p_x \\ a_{1y} + b_{1y} + f_1 (\cos(\alpha_1) \cos(\gamma_{f1}) + \sin(\alpha_1) \sin(\beta_1) \sin(\gamma_{f1})) + d_1 (\cos(\alpha_1) \cos(\gamma_{d1}) + \sin(\alpha_1) \sin(\beta_1) \sin(\gamma_{d1})) + e_{1y} - p_y \\ a_{1z} + b_{1z} + f_1 \cos(\beta_1) \sin(\gamma_{f1}) + d_1 \cos(\beta_1) \sin(\gamma_{d1}) + e_{1z} - p_z \\ a_{2x} + b_{2x} + f_2 (\cos(\alpha_2) \sin(\beta_2) \sin(\gamma_{f2}) - \cos(\gamma_{f2}) \sin(\alpha_2)) + d_2 (\cos(\alpha_2) \sin(\beta_2) \sin(\gamma_{d2}) - \cos(\gamma_{d2}) \sin(\alpha_2)) + e_{2x} - p_x \\ a_{2y} + b_{2y} + f_2 (\cos(\alpha_2) \cos(\gamma_{f2}) + \sin(\alpha_2) \sin(\beta_2) \sin(\gamma_{f2})) + d_2 (\cos(\alpha_2) \cos(\gamma_{d2}) + \sin(\alpha_2) \sin(\beta_2) \sin(\gamma_{d2})) + e_{2y} - p_y \\ a_{2z} + b_{2z} + f_2 \cos(\beta_2) \sin(\gamma_{f2}) + d_2 \cos(\beta_2) \sin(\gamma_{d2}) + e_{2z} - p_z \\ a_{3x} + b_{3x} + f_3 \sin(\beta_3) \sin(\gamma_{f3}) + d_3 \sin(\beta_3) \sin(\gamma_{d3}) + e_{3x} - p_x \\ a_{3y} + b_{3y} + f_3 \cos(\gamma_{f3}) + d_3 \cos(\gamma_{d3}) + e_{3y} - p_y \\ a_{3z} + b_{3z} + f_3 \cos(\beta_3) \sin(\gamma_{f3}) + d_3 \cos(\beta_3) \sin(\gamma_{d3}) + e_{3z} - p_z \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix}_{12 \times 1}$$

$$\mathbf{q} = [b_{1x} \quad b_{2x} \quad b_{3x} \quad \gamma_{f1} \quad \gamma_{d1} \quad \gamma_{f2} \quad \gamma_{d2} \quad \gamma_{f3} \quad \gamma_{d3} \quad p_x \quad p_y \quad p_z]^T$$

We find the Jacobian matrix by derivation of the constraint function $\Phi(\mathbf{q})$ with respect to vector of variables $\mathbf{q}$.

$$\begin{bmatrix} 0 & \cos(\gamma_{f1}) \sin(\beta_1) \cos(\alpha_1) \sin(\gamma_{f1}) - \cos(\gamma_{f1}) \sin(\alpha_1) & 0 & \cos(\gamma_{d1}) \sin(\beta_1) \cos(\alpha_1) \sin(\gamma_{d1}) - \cos(\gamma_{d1}) \sin(\alpha_1) & 1 & 0 & 0 \\ 0 & \sin(\gamma_{f1}) \sin(\beta_1) \cos(\alpha_1) \cos(\gamma_{f1}) + \sin(\alpha_1) \sin(\beta_1) \sin(\gamma_{f1}) & 0 & \sin(\gamma_{d1}) \sin(\beta_1) \cos(\alpha_1) \cos(\gamma_{d1}) + \sin(\alpha_1) \sin(\beta_1) \sin(\gamma_{d1}) & 0 & 1 & 0 \\ 0 & \sin(\beta_1) \cos(\gamma_{f1}) & 0 & \sin(\beta_1) \cos(\gamma_{d1}) & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \cos(\gamma_{f2}) \sin(\beta_2) \cos(\alpha_2) \sin(\gamma_{f2}) - \cos(\gamma_{f2}) \sin(\alpha_2) & 0 & 0 \\ 0 & 0 & 0 & 0 & \sin(\gamma_{f2}) \sin(\beta_2) \cos(\alpha_2) \cos(\gamma_{f2}) + \sin(\alpha_2) \sin(\beta_2) \sin(\gamma_{f2}) & 0 & 0 \\ 0 & 0 & 0 & 0 & \cos(\beta_2) \sin(\gamma_{f2}) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \cos(\gamma_{d2}) \sin(\beta_2) \cos(\alpha_2) \sin(\gamma_{d2}) - \cos(\gamma_{d2}) \sin(\alpha_2) & 0 \\ 0 & 0 & 0 & 0 & 0 & \sin(\gamma_{d2}) \sin(\beta_2) \cos(\alpha_2) \cos(\gamma_{d2}) + \sin(\alpha_2) \sin(\beta_2) \sin(\gamma_{d2}) & 0 \\ 0 & 0 & 0 & 0 & 0 & \cos(\beta_2) \sin(\gamma_{d2}) & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} 1 & 0 & 0 & f_1(\cos(\alpha_1)\sin(\beta_1)\cos(\gamma_{f1}) + \sin(\gamma_{f1})\sin(\alpha_1)) & d_1(\cos(\alpha_1)\sin(\beta_1)\cos(\gamma_{d1}) + \sin(\gamma_{d1})\sin(\alpha_1)) & 0 \\ 0 & 0 & 0 & f_1(\sin(\alpha_1)\sin(\beta_1)\cos(\gamma_{f1}) - \cos(\alpha_1)\sin(\gamma_{f1})) & d_1(\sin(\alpha_1)\sin(\beta_1)\cos(\gamma_{d1}) - \cos(\alpha_1)\sin(\gamma_{d1})) & 0 \\ 0 & 0 & 0 & f_1\cos(\beta_1)\cos(\gamma_{f1}) & d_1\cos(\beta_1)\cos(\gamma_{d1}) & 0 \\ 0 & 1 & 0 & 0 & 0 & f_2(\cos(\alpha_2)\sin(\beta_2)\cos(\gamma_{f2}) + \sin(\gamma_{f2})\sin(\alpha_2)) \\ 0 & 0 & 0 & 0 & 0 & f_2(\sin(\alpha_2)\sin(\beta_2)\cos(\gamma_{f2}) - \cos(\alpha_2)\sin(\gamma_{f2})) \\ 0 & 0 & 0 & 0 & 0 & f_2\cos(\beta_2)\cos(\gamma_{f2}) \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ d_2(\cos(\alpha_2)\sin(\beta_2)\cos(\gamma_{d2}) + \sin(\gamma_{d2})\sin(\alpha_2)) & 0 & 0 & -1 & 0 & 0 \\ d_2(\sin(\alpha_2)\sin(\beta_2)\cos(\gamma_{d2}) - \cos(\alpha_2)\sin(\gamma_{d2})) & 0 & 0 & 0 & -1 & 0 \\ d_2\cos(\beta_2)\cos(\gamma_{d2}) & 0 & 0 & 0 & 0 & -1 \\ 0 & f_3\sin(\beta_3)\cos(\gamma_{f3}) & d_3\sin(\beta_3)\cos(\gamma_{d3}) & -1 & 0 & 0 \\ 0 & -f_3\sin(\gamma_{f3}) & -d_3\sin(\gamma_{d3}) & 0 & -1 & 0 \\ 0 & f_3\cos(\beta_3)\cos(\gamma_{f3}) & d_3\cos(\beta_3)\cos(\gamma_{d3}) & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}_{12 \times 12}$$

We compute the time derivative of the Jacobian matrix,

$$\mathbf{\dot{J}} = \begin{bmatrix} 0 & 0 & 0 & f_1(\cos(\gamma_{f1})\sin(\alpha_1) - \cos(\alpha_1)\sin(\beta_1)\sin(\gamma_{f1}))\dot{\gamma}_{f1} & d_1(\cos(\gamma_{d1})\sin(\alpha_1) - \cos(\alpha_1)\sin(\beta_1)\sin(\gamma_{d1}))\dot{\gamma}_{d1} & 0 \\ 0 & 0 & 0 & -f_1(\sin(\alpha_1)\sin(\beta_1)\sin(\gamma_{f1}) + \cos(\alpha_1)\cos(\gamma_{f1}))\dot{\gamma}_{f1} & -d_1(\sin(\alpha_1)\sin(\beta_1)\sin(\gamma_{d1}) + \cos(\alpha_1)\cos(\gamma_{d1}))\dot{\gamma}_{d1} & 0 \\ 0 & 0 & 0 & -f_1\cos(\beta_1)\sin(\gamma_{f1})\dot{\gamma}_{f1} & -d_1\cos(\beta_1)\sin(\gamma_{d1})\dot{\gamma}_{d1} & 0 \\ 0 & 0 & 0 & 0 & 0 & f_2(\cos(\gamma_{f2})\sin(\alpha_2) - \cos(\alpha_2)\sin(\beta_2)\sin(\gamma_{f2}))\dot{\gamma}_{f2} \\ 0 & 0 & 0 & 0 & 0 & -f_2(\sin(\alpha_2)\sin(\beta_2)\sin(\gamma_{f2}) + \cos(\alpha_2)\cos(\gamma_{f2}))\dot{\gamma}_{f2} \\ 0 & 0 & 0 & 0 & 0 & -f_2\cos(\beta_2)\sin(\gamma_{f2})\dot{\gamma}_{f2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ d_2(\cos(\gamma_{d2})\sin(\alpha_2) - \cos(\alpha_2)\sin(\beta_2)\sin(\gamma_{d2}))\dot{\gamma}_{d2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -d_2(\sin(\alpha_2)\sin(\beta_2)\sin(\gamma_{d2}) + \cos(\alpha_2)\cos(\gamma_{d2}))\dot{\gamma}_{d2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -d_2\cos(\beta_2)\sin(\gamma_{d2})\dot{\gamma}_{d2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -f_3\sin(\beta_3)\sin(\gamma_{f3})\dot{\gamma}_{f3} & -d_3\sin(\beta_3)\sin(\gamma_{d3})\dot{\gamma}_{d3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -f_3\cos(\gamma_{f3})\dot{\gamma}_{f3} & -d_3\cos(\gamma_{f3})\dot{\gamma}_{d3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -f_3\cos(\beta_3)\sin(\gamma_{f3})\dot{\gamma}_{f3} & -d_3\cos(\beta_3)\sin(\gamma_{d3})\dot{\gamma}_{d3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{12 \times 12}$$

We compute the explicit time derivatives of the constraint function,

$$\dot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\dot{p}_x^0 - \ddot{p}_x^0 t \\ -\dot{p}_y^0 - \ddot{p}_y^0 t \\ -\dot{p}_z^0 - \ddot{p}_z^0 t \end{bmatrix}_{12 \times 1} \quad \ddot{\phi}_t = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -\ddot{p}_x^0 \\ -\ddot{p}_y^0 \\ -\ddot{p}_z^0 \end{bmatrix}_{12 \times 1}$$

Here is important to remark that each close loop equation, which is modeling each arm of the robot, can be solved independently from the other loop equations, but keep in mind that in this example the three system of equations are not similar. Here the system of equations modeling arms 1 and 2 are similar but the system of equation for arm 3 is different. The similarity was lost when the α rotation about "z" axis was set to zero for arm 3 and the rotation matrix $\mathbf{A}$ was simplified. If desired we can assume that the rotation angle α is different from zero leaving matrix $\mathbf{A}$ without simplification, hence leaving the system

of equation for arm 3 similar to the equations for arms 1 and 2. The angle α is later set to zero in the code.

Matlab code

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% This code solves the kinematic analysis of the linear tripteron robot with
% parallel and coplanar slider. The code is uses multibody kinematics based
% on the use of close loop equations
% a: location of the origin for the linear actuators
% b: vector from origin of the slider to R-joint of arm "f"
% f: length of the links of the first arm (slider to midjoint)
% d: length of the links of the second arm (midjoint to effector)
% e: location of the POI in the effector from the R-joints
% q: vector with the kinematic variables
% P: Matrix with the historic position
% V: Matrix with the historic velocities
% Acc: Matrix with the historic acceleration
% Fq: Vector with kinematic constraints
% J: Jacobian matrix
%
% the code uses ZYX rotation matrix (Tait-Bryan angles alpha,beta,gamma) to
% reduce the number of variables in the systema of equations.
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear all
clc
close all
% the GCS is located at one end of slider in the middle
% vectors a_i are used for positioning of the slider with respect the GCS
a1=[0;-40; 0];
a2=[0; 40; 0];
a3=[0; 0; 0];
%vectors from start of slider to the R-joint of link "f". "bx" is a variable
b1y=-7.33;
b1z=15.29;
b2y=7.33;
b2z=15.29;
b3y=12;
b3z=24.71;
%constant rotation angles to orientate the R-joint's rotation axes
%alpha is rotation about the "z" axis, beta is rotation about the "y" axis
alpha1=deg2rad(-45);
beta1=deg2rad(45);
alpha2=deg2rad(45);
beta2=deg2rad(45);
alpha3=deg2rad(0);
beta3=deg2rad(-45);
%constant vector "e" to located the POI in the effector with respect the
%R-joints
e1=[13.67; 13.33; 20.71];
e2=[13.67;-13.33; 20.71];
e3=[-8.64;-10.83; 20.71];
%length of parallelogram links
f=60;
d=60;
%simple driving constraints
px0=100; vx0=8; ax0=0;
py0=0; vy0=4; ay0=0;
pz0=20; vz0=4; az0=0;
%vector of unknowns
%q=[b1x b2x b3x gamma_f1 gamma_d1 gamma_f2 gamma_d2 gamma_f3 gamma_d3 px py pz ]
q=[34; 32; 160; deg2rad(90); deg2rad(10); deg2rad(80); deg2rad(170); deg2rad(20); deg2rad(150); 126; 0; 72 ];
%this initial guess for q is for the solution of effector at (0,0,0)
%solver initialization
counter=0;
ti=0;
tstep=0.1;
tf=10;

```

```

for t=ti:tstep:tf
    counter=counter+1;
    res=10;
    iter=1;
    while res>0.00001 && iter<100
        Fq=[a1(1)+q(1)+f*(cos(alpha1)*sin(beta1)*sin(q(4))-cos(q(4))*sin(alpha1))+d*(cos(alpha1)*sin(beta1)*sin(q(5))-cos(q(5))*sin(alpha1))+e1(1)-q(10);
            a1(2)+b1y+f*(cos(alpha1)*cos(q(4))+sin(alpha1)*sin(beta1)*sin(q(4)))+d*(cos(alpha1)*cos(q(5))+sin(alpha1)*sin(beta1)*sin(q(5)))+e1(2)-q(11);
            a1(3)+b1z+f*cos(beta1)*sin(q(4))+d*cos(beta1)*sin(q(5))+e1(3)-q(12);
            a2(1)+q(2)+f*(cos(alpha2)*sin(beta2)*sin(q(6))-cos(q(6))*sin(alpha2))+d*(cos(alpha2)*sin(beta2)*sin(q(7))-cos(q(7))*sin(alpha2))+e2(1)-q(10);
            a2(2)+b2y+f*(cos(alpha2)*cos(q(6))+sin(alpha2)*sin(beta2)*sin(q(6)))+d*(cos(alpha2)*cos(q(7))+sin(alpha2)*sin(beta2)*sin(q(7)))+e2(2)-q(11);
            a2(3)+b2z+f*cos(beta2)*sin(q(6))+d*cos(beta2)*sin(q(7))+e2(3)-q(12);
            a3(1)+q(3)+f*sin(beta3)*sin(q(8))+d*sin(beta3)*sin(q(9))+e3(1)-q(10);
            a3(2)+b3y+f*cos(q(8))+d*cos(q(9))+e3(2)-q(11);
            a3(3)+b3z+f*cos(beta3)*sin(q(8))+d*cos(beta3)*sin(q(9))+e3(3)-q(12);
            %driving constraints for a circular spiral
            q(10)-100-15*cos((2*pi/(tf/4))*t);
            q(11)-15*sin((2*pi/(tf/4))*t);
            q(12)-20-5*t;
            %simple driving constraints
            %q(10)-px0-vx0*t-0.5*ax0*t^2;
            %q(11)-py0-vy0*t-0.5*ay0*t^2;
            %q(12)-pz0-vz0*t-0.5*az0*t^2;
        ];
        J=[1,0,0,f*(cos(alpha1)*sin(beta1)*cos(q(4))+sin(q(4))*sin(alpha1)),d*(cos(alpha1)*sin(beta1)*cos(q(5))+sin(q(5))*sin(alpha1)), 0, 0, 0, 0, -1, 0, 0;
            0,0,0,f*(sin(alpha1)*sin(beta1)*cos(q(4))-cos(alpha1)*sin(q(4))),d*(sin(alpha1)*sin(beta1)*cos(q(5))-cos(alpha1)*sin(q(5))), 0, 0, 0, 0, 0,-1, 0;
            0,0,0,f*cos(beta1)*cos(q(4)), d*cos(beta1)*cos(q(5)), 0, 0, 0, 0, 0, 0, 0,-1;
            0,1,0,0, 0, f*(cos(alpha2)*sin(beta2)*cos(q(6))+sin(q(6))*sin(alpha2)), d*(cos(alpha2)*sin(beta2)*cos(q(7))+sin(q(7))*sin(alpha2)), 0, 0, -1, 0, 0;
            0,0,0,0, 0, f*(sin(alpha2)*sin(beta2)*cos(q(6))-cos(alpha2)*sin(q(6))), d*(sin(alpha2)*sin(beta2)*cos(q(7))-cos(alpha2)*sin(q(7))), 0, 0, 0,-1, 0;
            0,0,0,0, 0, f*cos(beta2)*cos(q(6)), d*cos(beta2)*cos(q(7)), 0, 0, 0, 0,-1;
            0,0,1,0, 0, 0, f*sin(beta3)*cos(q(8)), d*sin(beta3)*cos(q(9)), -1, 0, 0;
            0,0,0,0, 0, 0, -f*sin(q(8)), -d*sin(q(9)), 0,-1, 0;
            0,0,0,0, 0, 0, f*cos(beta3)*cos(q(8)), d*cos(beta3)*cos(q(9)), 0, 0,-1;
            0,0,0,0, 0, 0, 0, 0, 0, 1, 0, 0;
            0,0,0,0, 0, 0, 0, 0, 0, 0, 1, 0;
            0,0,0,0, 0, 0, 0, 0, 0, 0, 0, 1;
        ];
        qf = q - J\Fq;
        q = qf;
        res = norm(Fq);
        iter=iter+1;
    end
    if iter>=99
        disp(strcat("no convergency, number of iterations exceeded. frame=",num2str(counter)))
        break
    end
    P(:,counter)=qf;
    time(:,counter)=t;
%velocity analysis
    vi=-J[0; 0; 0; 0; 0; 0; 0; 0; 0; -vx0-ax0*t; -vy0-ay0*t; -vz0-az0*t];
    V(:,counter)=vi;
%acceleration analysis
    Jp=[0,0,0, f*vi(4)*cos(q(4))*sin(alpha1)-cos(alpha1)*sin(beta1)*sin(q(4)), d*vi(5)*cos(q(5))*sin(alpha1)-cos(alpha1)*sin(beta1)*sin(q(5)), 0, 0, 0, 0, 0, 0;
        0,0,0, -f*vi(4)*sin(alpha1)*sin(beta1)*sin(q(4))+cos(alpha1)*cos(q(4)), -d*vi(5)*sin(alpha1)*sin(beta1)*sin(q(5))+cos(alpha1)*cos(q(5)), 0, 0, 0, 0, 0, 0;
        0,0,0, -f*vi(4)*cos(beta1)*sin(q(4)), -d*vi(5)*cos(beta1)*sin(q(5)), 0, 0, 0, 0, 0, 0;
        0,0,0,0, 0, f*vi(6)*cos(q(6))*sin(alpha2)-cos(alpha2)*sin(beta2)*sin(q(6)), d*vi(7)*cos(q(7))*sin(alpha2)-cos(alpha2)*sin(beta2)*sin(q(7)), 0, 0, 0, 0, 0;
        0,0,0,0, 0, -f*vi(6)*sin(alpha2)*sin(beta2)*sin(q(6))+cos(alpha2)*cos(q(6)), -d*vi(7)*sin(alpha2)*sin(beta2)*sin(q(7))+cos(alpha2)*cos(q(7)), 0, 0, 0, 0, 0;
        0,0,0,0, 0, -f*vi(6)*cos(beta2)*sin(q(6)), -d*vi(7)*cos(beta2)*sin(q(7)), 0, 0, 0, 0, 0;
        0,0,0,0, 0, 0, -f*vi(8)*sin(beta3)*sin(q(8)), -d*vi(9)*sin(beta3)*sin(q(9)), 0, 0, 0;
        0,0,0,0, 0, 0, -f*vi(8)*cos(q(8)), -d*vi(9)*cos(q(9)), 0, 0, 0;
        0,0,0,0, 0, 0, 0, 0, -f*vi(8)*cos(beta3)*sin(q(8)), -d*vi(9)*cos(beta3)*sin(q(9)), 0, 0, 0;
        0,0,0,0, 0, 0, 0, 0, 0, 0, 0, 0;
        0,0,0,0, 0, 0, 0, 0, 0, 0, 0, 0;
        0,0,0,0, 0, 0, 0, 0, 0, 0, 0, 0;
    ];
    ai=J\(-Jp*vi-[0; 0; 0; 0; 0; 0; 0; 0; 0; -ax0; -az0; -az0]);
    Acc(:,counter)=ai;
end
%convert results to degrees
%Pdeg=zeros(size(P));
Pdeg(1:3,:)=P(1:3,:);
Pdeg(4:9,:)=rad2deg(P(4:9,:));
Pdeg(10:12,:)=P(10:12,:);

```

```

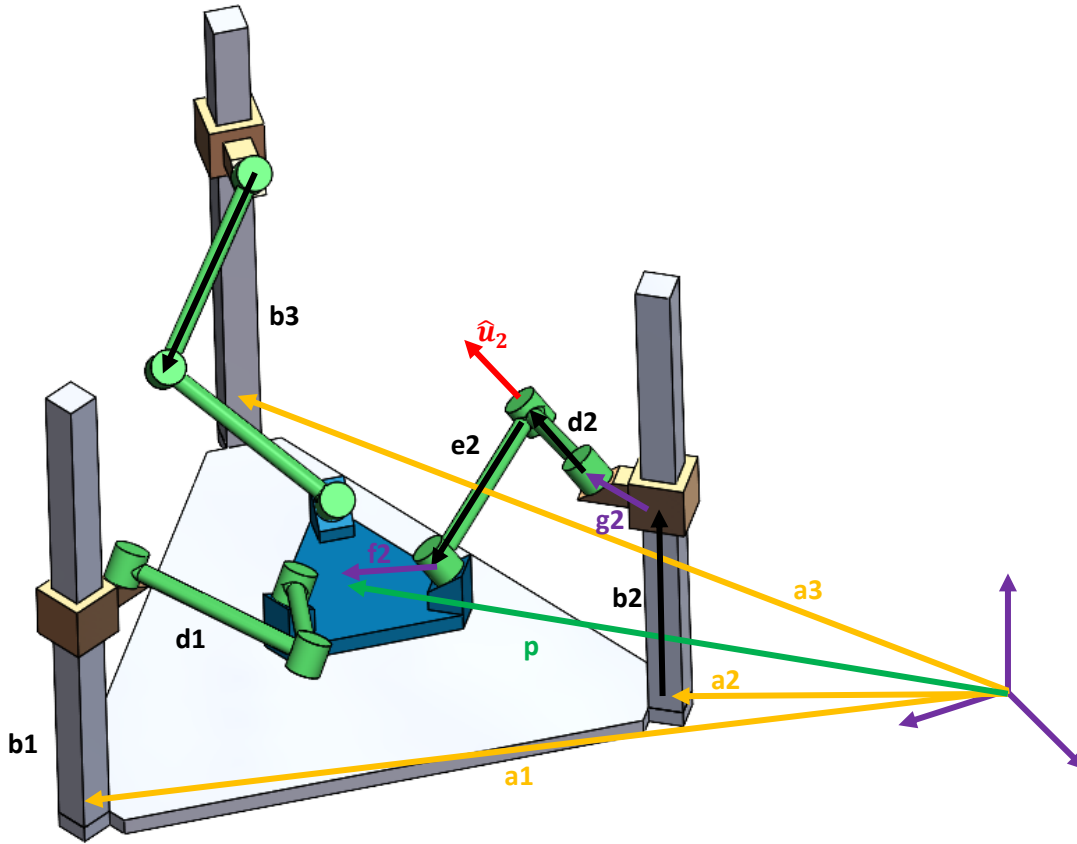
%animate the position solution
nSteps=size(P,2);
figure
pause
for i=1:nSteps
    cla
    O=[0; 0; 0];
    %first arm
    A=a1;
    B=a1+[P(1,i); b1y; b1z];
    F=B+Rot(alpha1,beta1,P(4,i))*[0;f;0];
    D=F+Rot(alpha1,beta1,P(5,i))*[0;d;0];
    E=D+e1;
    plot3([O(1) A(1)], [O(2) A(2)], [O(3) A(3)],'y')
    hold on
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)],'r')
    plot3([B(1) F(1)], [B(2) F(2)], [B(3) F(3)],'g')
    plot3([F(1) D(1)], [F(2) D(2)], [F(3) D(3)],'b')
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)],'m')
    %first slider
    Q=A+[250;0;0];
    plot3([A(1) Q(1)], [A(2) Q(2)], [A(3) Q(3)],'k')
    %second arm
    A=a2;
    B=a2+[P(2,i); b2y; b2z];
    F=B+Rot(alpha2,beta2,P(6,i))*[0;f;0];
    D=F+Rot(alpha2,beta2,P(7,i))*[0;d;0];
    E=D+e2;
    plot3([O(1) A(1)], [O(2) A(2)], [O(3) A(3)],'y')
    hold on
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)],'r')
    plot3([B(1) F(1)], [B(2) F(2)], [B(3) F(3)],'g')
    plot3([F(1) D(1)], [F(2) D(2)], [F(3) D(3)],'b')
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)],'m')
    %second slider
    Q=A+[250;0;0];
    plot3([A(1) Q(1)], [A(2) Q(2)], [A(3) Q(3)],'k')
    %third arm
    A=a3;
    B=a3+[P(3,i); b3y; b3z];
    F=B+Rot(alpha3,beta3,P(8,i))*[0;f;0];
    D=F+Rot(alpha3,beta3,P(9,i))*[0;d;0];
    E=D+e3;
    plot3([O(1) A(1)], [O(2) A(2)], [O(3) A(3)],'y')
    hold on
    plot3([A(1) B(1)], [A(2) B(2)], [A(3) B(3)],'r')
    plot3([B(1) F(1)], [B(2) F(2)], [B(3) F(3)],'g')
    plot3([F(1) D(1)], [F(2) D(2)], [F(3) D(3)],'b')
    plot3([D(1) E(1)], [D(2) E(2)], [D(3) E(3)],'m')
    %third slider
    Q=A+[250;0;0];
    plot3([A(1) Q(1)], [A(2) Q(2)], [A(3) Q(3)],'k')
    %POI
    plot3([O(1) P(10,i)], [O(2) P(11,i)], [O(3) P(12,i)], 'o')
    %effector
    X=P(10,i)+[-25,25,25,-25];
    Y=P(11,i)+[-16,-16,16,16];
    Z=P(12,i)+[0,0,0,0];
    fill3(X,Y,Z,'c')
    %trajectory
    plot3(P(10,1:i),P(11,1:i),P(12,1:i),'.','colon',[0.85 0.325 0.098])

    xlabel('x')
    ylabel('y')
    zlabel('z')
    title('animation of the position')
    axis equal
    axis([-10 260 -80 80 -10 100])
    view(-124.9665,28.2082)
    pause(0.1)
    grid on
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% this is an ZYX rotation matrix (Tait-Bryan angles alpha,beta,gamma)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function [A]=Rot(alpha,beta,gamma)
    A=[cos(alpha)*cos(beta), cos(alpha)*sin(beta)*sin(gamma)-cos(gamma)*sin(alpha), sin(alpha)*sin(gamma)+cos(alpha)*cos(gamma)*sin(beta);
        cos(beta)*sin(alpha), cos(alpha)*cos(gamma)+sin(alpha)*sin(beta)*sin(gamma), cos(gamma)*sin(alpha)*sin(beta)-cos(alpha)*sin(gamma);
        -sin(beta), cos(beta)*sin(gamma), cos(beta)*cos(gamma)];
end

```

12. Linear Tripteron – Parallel sliders



$$\Phi(\mathbf{q}) = \mathbf{0} = [\mathbf{a}_1 + \mathbf{b}_1 + \mathbf{g}_1 + \mathbf{d}_1 + \mathbf{e}_1 + \mathbf{f}_1 - \mathbf{p}]$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + 0 + g_{1x} + d \cos(\alpha_{d1}) + e \cos(\alpha_{e1}) + f_{1x} - p_x \\ a_{1y} + 0 + g_{1y} + d \cos(\beta_{d1}) + e \cos(\beta_{e1}) + f_{1y} - p_y \\ a_{1z} + b_1 + g_{1z} + d \cos(\gamma_{d1}) + e \cos(\gamma_{e1}) + f_{1z} - p_z \end{bmatrix}$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + 0 + g_{1x} + d \cos(\alpha_{d1}) + e \cos(\alpha_{e1}) + f_{1x} - p_x \\ a_{1y} + 0 + g_{1y} + d \cos(\beta_{d1}) + e \cos(\beta_{e1}) + f_{1y} - p_y \\ a_{1z} + b_1 + g_{1z} + d \cos(\gamma_{d1}) + e \cos(\gamma_{e1}) + f_{1z} - p_z \\ \cos^2(\alpha_{d1}) + \cos^2(\beta_{d1}) + \cos^2(\gamma_{d1}) - 1 \\ \cos^2(\alpha_{e1}) + \cos^2(\beta_{e1}) + \cos^2(\gamma_{e1}) - 1 \\ u_{1x} \cos(\alpha_{d1}) + u_{1y} \cos(\beta_{d1}) + u_{1z} \cos(\gamma_{d1}) \\ u_{1x} \cos(\alpha_{e1}) + u_{1y} \cos(\beta_{e1}) + u_{1z} \cos(\gamma_{e1}) \end{bmatrix}$$

$$\Phi(\mathbf{q}) = \mathbf{0} = \begin{bmatrix} a_{1x} + 0 + g_{1x} + d \cos(\alpha_{d1}) + e \cos(\alpha_{e1}) + f_{1x} - p_x \\ a_{1y} + 0 + g_{1y} + d \cos(\beta_{d1}) + e \cos(\beta_{e1}) + f_{1y} - p_y \\ a_{1z} + b_1 + g_{1z} + d \cos(\gamma_{d1}) + e \cos(\gamma_{e1}) + f_{1z} - p_z \\ \cos^2(\alpha_{d1}) + \cos^2(\beta_{d1}) + \cos^2(\gamma_{d1}) - 1 \\ \cos^2(\alpha_{e1}) + \cos^2(\beta_{e1}) + \cos^2(\gamma_{e1}) - 1 \\ u_{1x} \cos(\alpha_{d1}) + u_{1y} \cos(\beta_{d1}) + u_{1z} \cos(\gamma_{d1}) \\ u_{1x} \cos(\alpha_{e1}) + u_{1y} \cos(\beta_{e1}) + u_{1z} \cos(\gamma_{e1}) \\ p_x - p_x^0 - \dot{p}_x^0 t - \frac{1}{2} \ddot{p}_x^0 t^2 \\ p_y - p_y^0 - \dot{p}_y^0 t - \frac{1}{2} \ddot{p}_y^0 t^2 \\ p_z - p_z^0 - \dot{p}_z^0 t - \frac{1}{2} \ddot{p}_z^0 t^2 \end{bmatrix} \quad \mathbf{q} = \begin{bmatrix} b_1 \\ \alpha_{d1} \\ \beta_{d1} \\ \gamma_{d1} \\ \alpha_{e1} \\ \beta_{e1} \\ \gamma_{e1} \\ p_x \\ p_y \\ p_z \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} 0 & -d \sin(\alpha_{d1}) & 0 & 0 & -e \sin(\alpha_{e1}) & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -d \sin(\beta_{d1}) & 0 & 0 & -e \sin(\beta_{e1}) & 0 & 0 & -1 & 0 \\ 1 & 0 & 0 & -d \sin(\gamma_{d1}) & 0 & 0 & -e \sin(\gamma_{e1}) & 0 & 0 & -1 \\ 0 & -\sin(2\alpha_{d1}) & -\sin(2\beta_{d1}) & -\sin(2\gamma_{d1}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\sin(2\alpha_{e1}) & -\sin(2\beta_{e1}) & -\sin(2\gamma_{e1}) & 0 & 0 & 0 \\ 0 & -u_{1x} \sin(\alpha_{d1}) & -u_{1y} \sin(\beta_{d1}) & -u_{1z} \sin(\gamma_{d1}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -u_{1x} \sin(\alpha_{e1}) & -u_{1y} \sin(\beta_{e1}) & -u_{1z} \sin(\gamma_{e1}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\dot{\mathbf{J}} = \begin{bmatrix} 0 & -d \cos(\alpha_{d1}) \dot{\alpha}_{d1} & 0 & 0 & -e \cos(\alpha_{e1}) \dot{\alpha}_{e1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -d \cos(\beta_{d1}) \dot{\beta}_{d1} & 0 & 0 & -e \cos(\beta_{e1}) \dot{\beta}_{e1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -d \cos(\gamma_{d1}) \dot{\gamma}_{d1} & 0 & 0 & -e \cos(\gamma_{e1}) \dot{\gamma}_{e1} & 0 & 0 & 0 \\ 0 & -2\dot{\alpha}_{d1} \cos(2\alpha_{d1}) & -2\dot{\beta}_{d1} \cos(2\beta_{d1}) & -2\dot{\gamma}_{d1} \cos(2\gamma_{d1}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -2\dot{\alpha}_{e1} \cos(2\alpha_{e1}) & -2\dot{\beta}_{e1} \cos(2\beta_{e1}) & -2\dot{\gamma}_{e1} \cos(2\gamma_{e1}) & 0 & 0 & 0 \\ 0 & -\dot{\alpha}_{d1} u_{1x} \cos(\alpha_{d1}) & -\dot{\beta}_{d1} u_{1y} \cos(\beta_{d1}) & -\dot{\gamma}_{d1} u_{1z} \cos(\gamma_{d1}) & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\dot{\alpha}_{e1} u_{1x} \cos(\alpha_{e1}) & -\dot{\beta}_{e1} u_{1y} \cos(\beta_{e1}) & -\dot{\gamma}_{e1} u_{1z} \cos(\gamma_{e1}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$